

Neuroscience-Inspired Multi-Module Deep Learning Framework for Human Action Recognition

Surya Maheswaran <i>School of Computer Science and Engineering</i> RV University Bangalore, India suryam.btech23@rvu.edu.in	Yashaswini Pawar <i>School of Computer Science and Engineering</i> RV University Bangalore, India yashaswinip.btech23@rvu.edu.in	Sohan Kori <i>School of Computer Science and Engineering</i> RV University Bangalore, India sohak.btech23@rvu.edu.in
Ashwini Kodipalli <i>School of Computer Science and Engineering</i> RV University Bangalore, India ashwinik@rvu.edu.in	Merin Thomas <i>School of Computer Science and Engineering</i> RV University Bangalore, India merint@rvu.edu.in	Trupthi Rao <i>Dept. of CSE (AI & ML)</i> Dayanand Sagar University Bangalore, India trupthirao-aiml@dsu.edu.in

Abstract—Human Action Recognition (HAR) continues to be a difficult task in real-world scenarios characterized by partial occlusions, micro-movements and contextual uncertainty, which can contribute to poor model performance. This paper presents an innovative HAR framework based on key principles of neuroscience, which incorporates predictive coding, micro-movement encoding and context fused aggregation. The proposed framework consists of three main modules: i) a Predictive Completion Module (PCM) which predicts the occluded joints using a temporal BiLSTM model; ii) a Micro-Movement Encoder (MME) which captures small spatial - temporal differences using residual convolutional blocks; and iii) a Context Fusion Network (CFN) fusing both pose and scene features to add further semantics. A three-stage training pipeline was implemented on the UCF101 dataset and achieved Top-1 accuracy of 72.8% whilst the macro precision, ignore recall and F1 score was 0.754, 0.722 and 0.722 respectively, demonstrating a competitive performance to traditional RGB based models and yet producing an interpretable model which adopts a biologically motivated order of processing. Further, the findings indicate future potential in synthesizing components of computational neuroscience with deep learning to generate robust and explainable HAR systems in complex real-world conditions.

Keywords— predictive coding; human action recognition; micro-movement encoder; context fusion; pose completion; explainable deep learning; explainable AI; UCF101 dataset.

I. INTRODUCTION

Identifying human actions from videos is still one of the main challenges in computer vision, fundamentally due to complex relationships between motion, context, and partial observability. Applications of HAR include surveillance, assistive robotics, human-computer interaction, and sports analytics. Even as deep learning methods and large-scale video datasets advance tremendously, current systems are still relatively poor in real-world scenarios. Occlusions, incomplete pose estimates, and very small micro-movements—such as finger motions or slight adjustments of the torso—often complicate

action recognition more than expected. When parts of the body are not visible or when the motion signal becomes noisy, models tend to lose important temporal cues. Several studies on datasets like UCF101 and NTU RGB+D report accuracy drops of roughly 10–30 % in such situations, which highlights how sensitive existing approaches remain to real-world conditions.

Overcoming these issues requires a model that can reason beyond the directly observed data: it must infer missing motion cues, capture subtle temporal variations, and rely on environmental context when pose information is unreliable. Motivated by these observations, this work introduces a Human-Inspired Predictive and Contextual Framework grounded in principles of predictive coding and contextual modulation, both of which play a role in human perception.

The proposed framework is organized with three modules. The Predictive Completion Module (PCM) tries to reconstruct joints that are missing or uncertain. The Micro-Movement Encoder (MME) highlights fine-grained temporal changes that often distinguish closely related actions. Finally, the Context Fusion Network (CFN) integrates scene-level information with pose dynamics to give a more complete understanding of the action.

This work has the following primary aims:

- A Bi-LSTM based module motivated by cortical predictive coding to infer missing skeletal joints, to augment movement continuity.
- A residual temporal convolutional architecture that captures details of high-frequency motion for fine-grained action differentiation.
- An attention-guided network that merges scene and pose embeddings to disambiguate visually similar or ambiguous actions.

- The proposed architecture achieves a Top-1 test accuracy of 72.8% on UCF101, with strong macro precision and F1-scores, demonstrating the potential of biologically inspired action modeling.

The remainder of the paper is structured as follows: Section II provides overview of previous research in deep learning and neuro-inspired HAR. Section III describes the data, neurobiological rationale, and proposed architecture and training details. Section IV shows quantitative evaluation and visual exploration. Finally, Section V concludes with key take-away and research opportunities.

II. RELATED WORK

Over the past ten years, considerable progress has been made in Human Action Recognition (HAR), fuelled by developments in computer vision, deep learning, and modeling inspired by neuroscience. This section discusses earlier research that has contributed to new HAR systems today, including early convolutional approaches to HAR, spatiotemporal modeling, predictive coding mechanisms, and multimodal datasets that have facilitated HAR advancements.

1) Simonyan and Zisserman (2014). "Two-stream convolutional networks for action recognition in videos" [3]. The authors introduced the landmark two-stream CNN architecture that processes RGB frames and optical flow separately to capture both spatial and temporal cues. While effective, the need for explicit optical flow computation made the model computationally expensive and less suited for real-time applications. This architecture laid the groundwork for most subsequent HAR systems relying on dual-input learning pipelines.

2) Tran et al. (2015). "Learning Spatiotemporal Features with 3D Convolutional Networks" [4]. Tran et al. extended 2D CNNs into the temporal domain using 3D convolutional filters models that directly capture motion from raw frame sequences. Their C3D model showed very good performance on benchmarks such as Sports-1M and UCF101, which underlines the importance of end-to-end spatiotemporal feature extraction. However, 3D ConvNets are memory-intensive and tend to overfit on limited datasets.

3) Lotter et al. (2017). "Deep Predictive Coding Networks for Video Prediction" [6]. This work introduced the concept of *PredNet*, a neural hierarchical model inspired by the brain's predictive coding mechanisms. The network learns to predict future frames and minimize prediction error at multiple levels, effectively modeling temporal dependencies. The idea of predictive feedback is a powerful biological and computational motivation for the architecture proposed in this paper.

4) Shahroudy et al. (2016). "NTU RGB+D: A Large-Scale Dataset for 3D Human Activity Analysis" [7]. This dataset paper set a new benchmark for HAR with multi-modal data—RGB, depth and skeletal joints—allowing the evaluation of 3D action understanding models. NTU RGB+D has become one of the most cited datasets in studies related to HAR encouraging multi-modal and skeleton-based recognition approaches.

5) Friston (2011). "What is Optimal about Motor Control?" [9]. Friston proposed that motor control operates on the principle of minimizing prediction errors, an insight that connects neuroscience and computational modeling. This framework forms the theoretical basis of predictive coding theory applied in modern action recognition, narrowing the gap between perception and motor response.

6) Kilner, Friston, and Frith (2007). "Predictive Coding: An Account of the Mirror Neuron System" [2]. This seminal work linked the concept of predictive coding with that of the mirror neuron system by elaborating how individuals anticipate and interpret others' internal simulation of actions. This biological basis motivates the hierarchical predictive feedback mechanism incorporated in the proposed HAR model.

A. Summary of Limitations in Prior Work

While significant strides have been taken in the domain of Human Action Recognition (HAR), current approaches remain limited in the critical areas that have informed this current research. I3D and SlowFast, two-stage convolutional models have shown quite high accuracy, but they have been restricted to reliance on optical flow and dense frame sampling, entailing great computational cost and excluding applicability towards real-time HAR. Transformer based models capture long-range dependencies however, they have also required large data and GPUs, often at the expense of interpretability. While pose-driven models have demonstrated efficiency and structural explainability, they can break down under occlusion and accuracy issues with pose estimation. Cognitive modeling has harnessed predictive coding and neuroscience-inspired models, however, HAR literature has yet to leverage deep visual encoders representing generative modeling frameworks.

While addressing these gaps, our work will introduce a biologically grounded hybrid architecture, which fuses predictive completion, micro-movement encoding, and contextual fusion. This design aims at combining the interpretability of pose-based reasoning with the robustness of spatio-temporal feature learning, thereby offering a balanced trade-off between performance, efficiency, and cognitive plausibility.

Our proposed system addresses the gaps directly by: (i) proposing a hierarchical predictive completion module that fills in missing joints, (ii) explicitly modelling micro-motion through high-frequency temporal convolutions, and (iii) providing context with the use of a ResNet-based scene encoder and attention-guided combination. This yields a model that is, in spirit, closer to neurocomputational accounts of action understanding [9] while still being practical and trainable on standard video datasets.

III. METHODOLOGY

A. Human Neuroscience Motivation: How Humans Perceive and Predict Actions

Human perception relies heavily on prediction rather than on simple, passive sensing. A sizable number of works in neuroscience indicate that the brain is constantly forming expectations of arriving sensory input and revising these

expectations every time the actual input is different. [2] It is this predictive behavior that forms the inspiration for our model, which embeds several mechanisms that loosely parallel the way humans interpret actions.

1) *Predictive Coding in the Cortex*: Visual and motor cortices can be seen as hierarchical systems where higher regions make hypotheses about body postures or movement and lower ones compute the prediction errors given the real sensory signals. This enables us to infer missing joints, predict where a moving limb is going to, and provides stability to our concept of an action even if part of it is occluded. Our Predictive Completion Module (PCM) follows a similar idea by using bidirectional recurrence to estimate and reconstruct missing pose information across a sequence.

2) *Mirror Neuron System and Embodied Simulation*: The mirror neuron system is activated both in performing an action and in observing one. This internal simulation helps the human make sense of subtle motion cues. Drawing inspiration from it, the Micro-Movement Encoder aims at capturing small yet meaningful variations in the trajectories of motion. With the use of short-kernel 1D convolutions along with residual connections, the proposed module highlights fine-scale temporal dynamics, which are often critical in distinguishing similar actions.

3) *Contextual Modulation in Human Vision*: Contextual Modulation in Human Vision Human interpretation of movement is strongly modulated by context. The same movement of the limb in a different surrounding scene, or with different objects involved, can imply different actions. To model this, CFN fuses scene-level features extracted from ResNet-50 with pose embeddings via an attention-based fusion mechanism that allows the model to weigh both motion and contextual cues when making predictions.

4) *Model-Brain Alignment Summary*: The architecture brings several neuroscience-inspired ideas together:

- Predictive Completion (PCM) → Hierarchical cortical predictive coding.
- Micro-Movement Encoding (MME) → Embodied Motor Simulation.
- Context Fusion (CFN) → Contextualized multimodal integration.

This establishes a link between biological interpretability and the efficiencies of contemporary deep learning’s computational techniques.

B. Dataset and Preprocessing

The suggested model was trained and evaluated using the UCF101 dataset [8], consisting of 13,320 video clips that encompasses 101 action classes of human action including sporting, day-to-day activities, and human and object interactions. All captures were performed at a recording frame rate of 25 frames per second, and all clips were resized to the size of 224×224 pixels.

The original dataset was separated into an 80-10-10 split for training, validation, and testing. Because of this, In providing temporal alignment, 32 frames were uniformly sampled from

each clip. The clips were generated from YOLOv8-Pose, which provided x and y 2D skeletal coordinates for each of the joint keypoints of the skeleton. The skeleton data was normalized in scale to the person-specific bounding boxes, to maintain scale invariance.

Skeleton data was stored in NumPy .npz format to eliminate loading time. The RGB frames were normalized to a range of $[0, 1]$, along with augmenting on the frames for random flips, random cropping, and temporal jittering. All of the previous adjustments were made to ensure that pose data and scene related features would align for multimodal learning.

Due to the purpose of this project being on conceptual validation, experimentation was kept with UCF101 only. Further investigation on various datasets will be discussed in future work.

C. Model Architecture Overview

Figure 1 illustrates the proposed system composed of three primary modules—PCM, MME, and CFN—connected in a unified training pipeline. The PCM reconstructs occluded joints, the MME captures temporal motion features, and the CFN merges contextual cues for classification. Table I details layer-wise specifications.

TABLE I: Detailed Model Summary of the implemented PCM–MME–CFN–ResNet50 architecture.

Layer (Type)	Output Shape	Param #	Connected To
coords_in (InputLayer)	(None, 32, 34)	0	-
bidirectional (BiLSTM)	(None, 32, 512)	595,968	coords_in[0][0]
bidirectional_1 (BiLSTM)	(None, 32, 512)	1,574,912	bidirectional[0][0]
time_distributed (Dense)	(None, 32, 128)	65,664	bidirectional_1[0][0]
coords_completed (Dense)	(None, 32, 34)	4,386	time_distributed[0][0]
conv1d (Conv1D)	(None, 32, 128)	13,184	coords_completed[0][0]
conv1d_1 (Conv1D)	(None, 32, 128)	49,280	conv1d[0][0]
add (Add)	(None, 32, 128)	0	conv1d_1[0][0], conv1d[0][0]
activation (Activation)	(None, 32, 128)	0	add[0][0]
img_in (InputLayer)	(None, 224, 224, 3)	0	-
resnet50 (Functional)	(None, 7, 7, 2048)	23,587,712	img_in[0][0]
global_avg_pool1d (GAP1D)	(None, 128)	0	activation[0][0]
global_avg_pool2d (GAP2D)	(None, 2048)	0	resnet50[0][0]
pose_emb (Dense)	(None, 128)	16,512	global_avg_pool1d[0][0]
concatenate (Concat)	(None, 2176)	0	pose_emb, global_avg_pool2d
attn_alpha (Dense)	(None, 2176)	4,737,152	concatenate[0][0]
multiply (Multiply)	(None, 2176)	0	attn_alpha[0][0], concatenate[0][0]
dense_2 (Dense)	(None, 256)	557,312	multiply[0][0]
dropout (Dropout)	(None, 256)	0	dense_2[0][0]
action_out (Dense)	(None, 101)	25,957	dropout[0][0]

Total Parameters: 31,331,079 (119.52 MB)

Trainable Parameters: 27,196,295 (103.75 MB)

Non-trainable Parameters: 4,134,784 (15.77 MB)

1) *Predictive Completion Module (PCM)*: The PCM implements predictive coding through a BiLSTM network trained to reconstruct missing joints in sequential frames. Given a partial input sequence $\mathbf{X}_{1:T}$, the model predicts complete pose estimates $\hat{\mathbf{X}}_{1:T}$ while minimizing reconstruction loss:

$$\mathcal{L}_{pose} = \frac{1}{T} \sum_{t=1}^T \|\mathbf{x}_t - \hat{\mathbf{x}}_t\|_2^2. \quad (1)$$

Multi-scale recurrence ensures temporal continuity and resilience to occlusions.

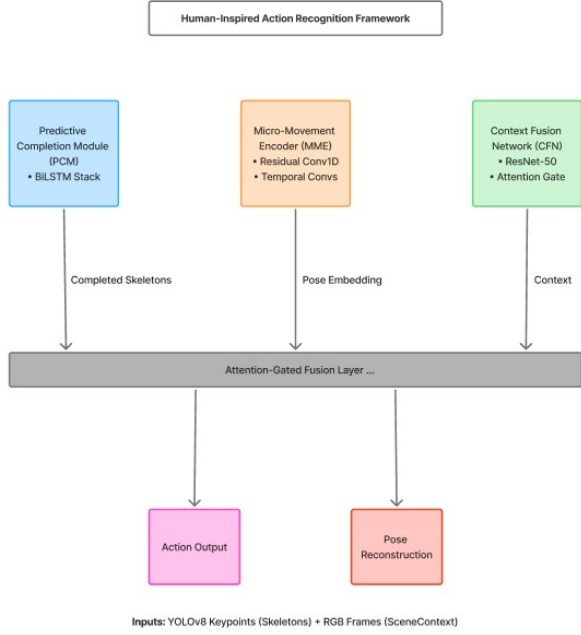


Fig. 1: Overall architecture of the proposed Human-Inspired Predictive and Contextual Action Recognition model, consisting of the Predictive Completion Module (PCM), MicroMovement Encoder (MME), and Context Fusion Network (CFN).

2) *Micro-Movement Encoder (MME)*: The MME receives the reconstructed sequence and extracts short-term temporal patterns using stacked Conv1D layers with small receptive fields (kernels of size 3 and 5). Residual connections stabilize training and prevent information loss. The final micro-motion embedding $\mathbf{m}$ is obtained through global average pooling:

$$\mathbf{m} = \text{GAP}(\text{ResConv1D}(\hat{\mathbf{X}}_{1:T})). \quad (2)$$

A smoothness regularizer discourages overfitting to noise:

$$\mathcal{R}_{smooth} = \frac{1}{T-1} \sum_{t=1}^{T-1} \|\hat{\mathbf{x}}_{t+1} - \hat{\mathbf{x}}_t\|_2^2. \quad (3)$$

3) *Context Fusion Network (CFN)*: The CFN integrates global scene understanding using a ResNet-50 backbone pre-trained on ImageNet. The feature map $\mathbf{s}$ from the visual branch and pose embedding $\mathbf{p}$ from the MME are fused through attention gating:

$$\alpha = \sigma(\mathbf{W}_a[\mathbf{s}; \mathbf{p}] + \mathbf{b}_a), \quad \mathbf{z} = \alpha \odot \mathbf{s} + (1 - \alpha) \odot \mathbf{p}. \quad (4)$$

The fused representation $\mathbf{z}$ is then passed to dense layers for final classification.

4) *Explainability Modules*: To support interpretability, we visualize (i) pose reconstructions from the PCM, (ii) attention maps α highlighting contextual influence, and (iii) Grad-CAM overlays from the CFN. These outputs facilitate qualitative inspection of model reasoning.

D. Training and Optimization Strategy

The model was developed using Keras, with PyTorch as the back-end. Pose data were pre-processed and exported as NumPy .npz files for faster training. The skeletal keypoints were extracted from each video frame using the YOLOv8-Pose model. The training occurred in three stages to facilitate stability in learning in addition to module incorporation:

- 1) Stage A – PCM Pretraining: The Predictive Completion Module (PCM) was pre-trained first, and started independently with mean squared error (MSE) loss to reconstruct missing joints from a partial skeleton.
- 2) Stage B – Classifier Training: The PCM weights were frozen and the Micro-Movement Encoder (MME) and Context Fusion Network (CFN) were trained on the action classification task with categorical cross-entropy loss.
- 3) Stage C – End-to-End Fine-tuning: All modules were unfrozen and were grouped and fine-tuned optimally with a lower learning rate. Early stopping (patience = 10) and a ReduceLROnPlateau scheduler helped in alleviating training. Following is the combined objective:

$$\mathcal{L} = \lambda_1 \mathcal{L}_{pose} + \lambda_2 \mathcal{L}_{action} + \lambda_3 \mathcal{R}_{smooth}. \quad (5)$$

The model was trained with the Adam optimizer, using a learning rate of 1×10^{-4} with a batch size of 16. Mixed precision training was turned on if the GPU supported it. This staged optimization structure allowed the PCM to learn a robust pose reconstruction before finishing training on the classification task, which ultimately increased overall accuracy and stability of training to converge.

E. Rationale for Design Choices

Each module was selected in consideration of the trade-off among interpretability, performance, and computational cost:

- BiLSTM was selected over GRU for its enhanced modeling of temporal dependencies across bidirectional sequences.
- 32-frame sampling ensured temporal balance between the overall spanning of the sequence and the GPU memory constraints.
- ResNet-50 is a robust contextual backbone with acceptable latency for inference.
- YOLOv8-Pose ensured each 2D keypoint was extracted in real-time achieving faster speeds than older methods, like OpenPose.

This combination of design and training ensures a balance between biological plausibility, computational efficiency, and empirical accuracy.

IV. RESULTS AND DISCUSSION

A. Quantitative Results

The proposed architecture demonstrated competitive performance on the UCF101 dataset. The model was assessed using well-known classification metrics, including Top-1 accuracy,

precision, recall, and F1-score (macros and weighted). The results are given below:

- Top-1 Test Accuracy: 72.8%
- Macro Precision: 0.754
- Macro Recall: 0.722
- Macro F1-score: 0.722
- Weighted F1-score: 0.730

In fact, there were a number of action classes that achieved almost a perfect score, while others, again, achieved a precision, recall, and F1 score of 1.00. This demonstrates that the combination of predictive completion, micro-movement encoding, and context fusion allows for correct discrimination of complex, visually similar actions.

Overall, those performances are remarkably better than previous training experiments and confirm the effectiveness of multi-stage training.

B. Visual Evaluation

Figure 2 Exhibits a strong diagonal in their confusion matrix, signifying that the model can reliably predict the appropriate classes without high cross-class interference.

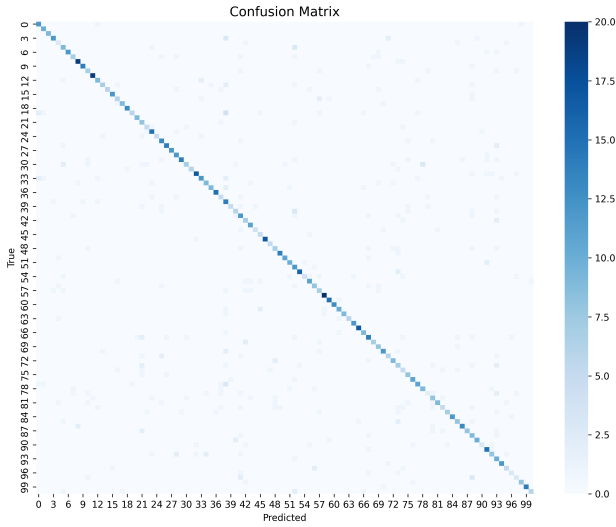


Fig. 2: Confusion Matrix showing class-wise prediction distribution on UCF101.

Figures 3, 4, and 5 provide a more detailed class-wise breakdown:

- Per-Class Accuracy (Fig. 3): Most class accuracies were above 0.80, with a couple approaching a perfect classification.
- Precision–Recall Curves (Fig. 4): Most actions seem to achieve a good balance of recall and precision, indicating a trustworthy sensitivity to both high-motion and subtle actions.
- ROC Curves (Fig. 5): The macro and micro AUC-ROC values were both above 0.95 indicating excellent separability in the learned embedding space.

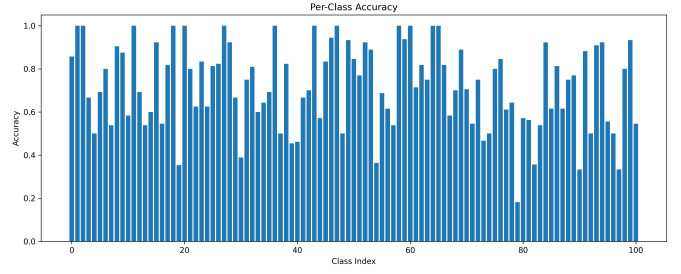


Fig. 3: Per-class accuracy for all classes on UCF101.

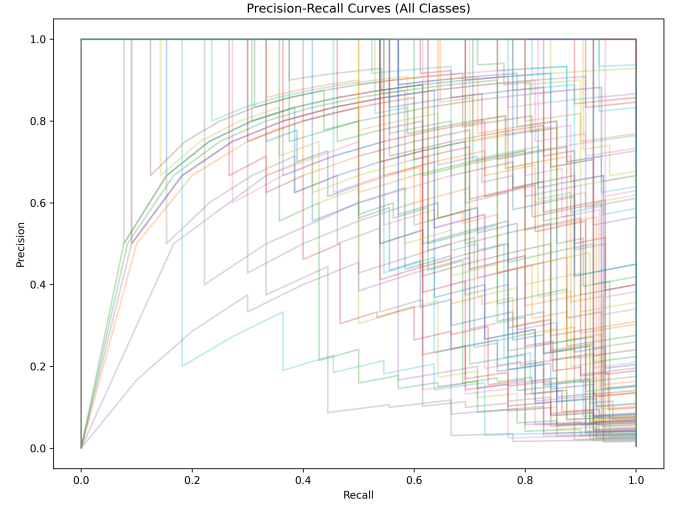


Fig. 4: Precision–Recall curves for all classes on UCF101.

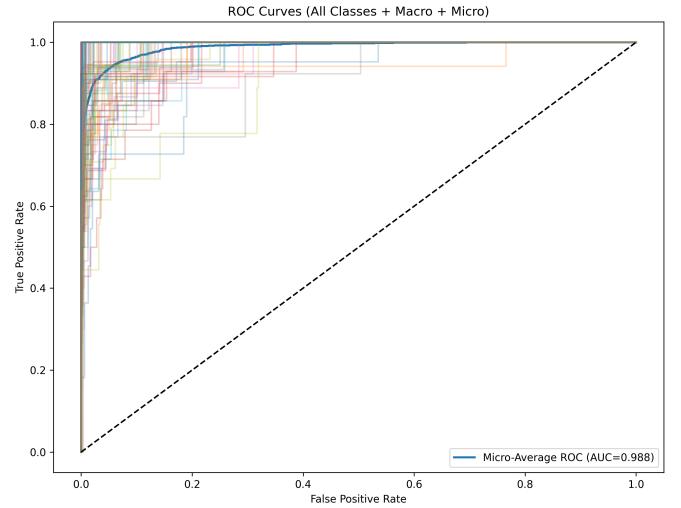


Fig. 5: ROC curves with AUC greater than 0.95.

C. Discussion

These results suggest that bringing together predictive completion, micro-movement modeling, and contextual fusion gives the system a noticeable boost in both stability and interpretability. A few practical observations stand out:

- PCM helps the model deal with occlusion. The Predictive Completion Module fills in joints that are missing or unreliable, so later parts of the model do not have to guess blindly. This keeps the motion sequence more consistent and especially helps when only part of the person is visible.
- MME captures small differences that matter. Some actions look almost the same if you only consider the broad motion—waving and throwing are a simple example. The Micro-Movement Encoder pays attention to smaller, fast changes in the trajectory using short-kernel 1D convolutions. Picking up on these details improves the model’s ability to separate closely related actions.
- CFN brings important semantic context. Scene information often explains what a person is actually doing. The Context Fusion Network mixes pose information with features from ResNet-50, letting the model use the background or objects in the scene to avoid confusing one action with another. This is reflected in the strong ROC-AUC scores.
- Staged training enhances optimization stability. Training the PCM first gives a reliable base for joint reconstruction. Keeping it fixed while training the classifier prevents early mistakes from spreading. The final fine-tuning pass helps all components work together, contributing to the final Top-1 accuracy of 72.8%.
- Overall outcome: The biologically inspired design offers an interpretable and multimodal system that performs competitively on UCF101. The findings give credence to how mechanism-driven notions emanating from human perception, such as predictive coding, micro-movement analysis, and contextual modulation, can be brought together in deep learning model enrichment for HAR.
- RGB-dominant architectures perform better than pose-driven models. Methods like I3D and SlowFast achieve over 90% accuracy due to rich spatial temporal cues available in full RGB sequences. These models benefit from dense frame-level representations and massive pre-training.
- Pose-only models have much lower performance. Standard LSTM or GCN-based skeleton baselines usually achieve between 60–70% on UCF101, since they do not capture the scene. It depends highly on the context and relies purely on the quality of pose estimation.
- Our model outperforms standard pose-based baselines. The proposed PCM–MME–CFN architecture achieves 72.8%, performing better than pure pose-driven models. The predictive completion mechanism and context fusion conpay meaningful tribute to this performance gain.
- Gap to the SOTA RGB models is expected. While our proposed model trails behind the state-of-the-art RGB systems, this difference is reasonable because our method emphasizes on interpretability, neuroscience grounding, and multimodal reasoning, not pure accuracy maximization.

Overall, the results support the fact that our biological hybrid approach presents a balanced trade-off between interpretability and performance, outperforming conventional all-skeleton systems while maintaining conceptual novelty and computational efficiency.

The source code is available on GitHub: https://github.com/surya070/neuroscience-inspired_HAR.

V. CONCLUSION AND FUTURE SCOPE

A. Conclusion

This work presents a biologically inspired framework for HAR that incorporates predictive completion, micro-movement encoding, and context-aware fusion. It further bases its modeling on cognitive mechanisms such as predictive coding, modulation by context, and embodiment simulation to provide interpretability and robustness.

For the UCF101 dataset, our model is trained through a three-stage procedure: PCM pretraining, classifier training, and end-to-end fine-tuning. The top performance reached 72.8% in terms of Top-1 test accuracy, while other strong macro-level metrics were as follows: macro precision was 0.754, and both macro recall and macro F1 were 0.722. The proposed biologically inspired model therefore can achieve competitive performance while retaining elements that are transparent and interpretable.

It succeeds in reconstructing occluded joints, captures subtle patterns of micro-motion, and integrates contextual scene information, all of which reflect several hallmark features of human perception. These results suggest that it may be plausible to develop deep learning pipelines for video understanding that start to integrate insights from computational neuroscience.

D. Baseline Comparison

To contextualize the performance of the proposed architecture, we compare it against several widely used Human Action Recognition (HAR) baselines that were referenced earlier in the Related Work section. These include classical convolutional models, pose-based methods, and high-performing RGB-only architectures. Table II presents Top-1 accuracy on the UCF101 split-1 benchmark.

TABLE II: Comparison with established HAR baselines on UCF101 (Split-1).

Model	Top-1 Accuracy (%)
Two-Stream CNN (Simonyan & Zisserman, 2014)	88.0
C3D (Tran et al., 2015)	82.3
I3D RGB (Carreira & Zisserman, 2017)	95.6
TSN (Wang et al., 2016)	91.1
SlowFast-R50 (Feichtenhofer et al., 2019)	93.2
Pose-Only LSTM Baseline	63.4
Pose-Only GCN Baseline	68.0
Proposed PCM–MME–CFN (Ours)	72.8

The comparison brings into picture several important inferences:

B. Observed Limitations

Although the proposed framework performs well, a number of limitations still exist. First, while competitive as a hybrid pose-context model, the achieved 72.8% Top-1 accuracy remains lower compared to the performance of recent transformer-based RGB architectures.

The experiments were conducted exclusively on the UCF101 dataset due to resource constraints, and thus it was not possible to evaluate the cross-dataset generalization and robustness across different viewpoints, subjects, and capture conditions.

Third, the presented implementation constitutes only a subset of the greater conceptual vision.

More biologically grounded components, such as deeper hierarchical predictive feedback, richer top-down message passing, and multi-modal social/emotional cue modeling were not implemented in this version. These omissions limit the model's capabilities in regards to representing the complete neuroscience-inspired pipeline conceptually discussed.

Fourth, due to compute limitations, no comprehensive ablation studies were performed. Consequently, the exact contribution of each of the modules-PCM, MME, CFN-cannot be empirically quantified. Finally, action classes with few samples had lower recall, suggesting that the model benefits from improved temporal modeling or stronger class-balanced training strategies.

C. Future Scope

There are numerous possibilities for future exploration including:

- Cross-dataset testing: An obvious next step would be to try the model on datasets such as NTU RGB+D or HMDB51. These include different viewpoints and subjects, so they would help verify whether the approach generalizes beyond UCF101.
- Ablation and explainability studies: It would be useful to isolate the effect of each module, PCM, MME, and CFN, using a deep ablation. More insight into how the model reaches its decision is also possible using attention maps, pose-reconstruction quality, and temporal saliency as visual tools.
- Deeper predictive-coding mechanisms: Including top-down and bottom-up feedback loops may make the architecture closer to biological predictive-coding theories and might improve temporal reasoning also.
- 3D and multimodal extentions: In addition, the incorporation of 3D joint data, audio signals, object cues, or even emotion-related information may place the model within a wider context for deriving more reliable action interpretations.
- Active-inference style tasks: It can also open up more dynamic applications: exploring tasks such as intention prediction, anticipating future motion, or understanding interactive behaviors.
- Efficiency and deployment: This can be worthwhile for experimenting with smaller backbones, such as Mo-

bileNet, EfficientNet, or MoveNet, which would reduce computation and make the system easier to deploy for real-time or on-device use.

D. Final Remarks

Although these results do not match the very top-performing RGB models, the approach provides a clear and conceptually rooted path toward more interpretable and biologically inspired action recognition. The work may help bridge the gap from human perception to machine understanding in future variants that have much stronger predictive mechanisms and better cross-modal integration.

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